Building Evidence Grounded AI Discovery with Koha DSpace and VuFind

An implementation case study with an illustrative paired pilot

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Abstract

A library discovery service may identify relevant books without having access to their full text. Adding generated answers therefore requires care about what each source can support. This case study describes a prototype connecting Koha catalogue records and DSpace repository material through VuFind, with an optional answer service using Qdrant, FastAPI and selectable hosted GPT or local Llama providers. Checks on 29 October 2026 recorded 200 Koha bibliographic records, 20 archived DSpace items with 20 linked PDFs, 220 VuFind records and 817 Qdrant points. An illustrative pilot sent the same saved evidence from four sources to both providers. The hosted answer used numerical citations and 53 words; the local answer used nonconforming labels, 81 words and described a journal article as a book. The saved repository passages were located on their original PDF pages, and selected catalogue fields and repository identifiers were reconciled. A subsequent open-label author assessment rated relevance and completeness at 4 of 4 for the hosted answer and 3 of 4 for the local answer. These observations identify implementation issues rather than establish comparative model performance. The case shows the value of preserving source identities, separating citation-format checks from claim support, and retaining failed answers for inspection. Independent evaluation and collection-wide reconciliation remain future work.

Keywords: library discovery; retrieval-augmented generation; Koha; DSpace; VuFind; Qdrant; source attribution

1 Introduction

A patron asking about a subject may need several kinds of material: a book from the catalogue, an article from a repository, or a brief explanation supported by those resources. A shared discovery interface can bring the records together. A generated answer adds a different task, because it must distinguish what the library owns from what the available text actually says.

That distinction shaped this project. A Koha record can establish a title, contributor, subject heading or contents note. It cannot establish every argument made in an unread book. A repository PDF may provide a passage that supports a more specific statement, provided that the passage is extracted correctly and linked to the right document and page. An answer service that treats both kinds of evidence alike risks giving catalogue descriptions the authority of full-text evidence.

I developed a prototype in which Koha and DSpace supply material to VuFind for discovery and to a separate retrieval-augmented generation service for optional answers. The service uses an existing language model at answer time; it does not train a new model on the catalogue. This follows the general retrieval-augmented approach described by Lewis et al. (2020), while retaining the existing library systems as the sources of records and documents.

The article examines how these components were connected and what could be checked in a small implementation. It extends the earlier project preprint (Jose, 2026a) with source-system count checks, a paired

pilot using saved evidence, original-source verification and a subsequent author assessment. Its purpose is to document an implementation and the problems encountered. A broad comparison of model quality is outside the evidence collected here.

2 Related work

Library applications of semantic retrieval and conversational search provide precedents for this work. Fitch (2023) described semantic search and cited question answering over newspaper content from the National Library of Australia's Trove repository. That implementation showed how an answer interface could retain links to library material while moving beyond a conventional list of keyword matches.

Mukhopadhyay (2025) demonstrated conversational library search using open-source RAG components, including LlamaIndex, PrivateGPT, Qdrant, Ollama and a Gradio interface. The present case uses custom application logic exposed through FastAPI and integrated with VuFind. This is an implementation choice suited to the prototype, rather than evidence that a custom service is inherently better than a framework-based alternative.

Evaluation also needs to separate retrieval from generation. Es et al. (2024) distinguish retrieved-context quality, faithfulness and answer quality in the RAGAs framework. That distinction is useful here: a correct source number does not show that the linked text supports a statement. The pilot therefore preserved the supplied evidence and original answers so that format compliance and substantive support could be examined separately. The literature reviewed here establishes relevant precedents; it is not an exhaustive survey or a claim of first implementation.

3 Case study setting and evidence

The prototype was developed on a Google Cloud virtual machine for a small research collection. The deployment inventory dated 5 October 2026 records Debian 12, four virtual CPUs, 16 GB RAM and a 100 GB disk. It lists Koha 26.05.03-1, DSpace 10.1, VuFind 11.1.0, Qdrant 1.19.1 and FastAPI 0.142.2. Later terminal output records Ollama 0.35.0. These values describe the documented deployment; hardware utilisation and memory peaks were not measured during the pilot.

Read-only checks on 29 October counted records in each source system and index (Table 1). The checks took place sequentially between 20:08 and 20:22 UTC. VuFind's collection facets returned 200 Catalog records and 20 DSpace records. The 817 Qdrant points represent stored evidence entries, not 817 separate documents. Their relationship to every source record has not yet been reconciled.

Table 1. Collection and index counts on 29 October 2026.

Measure	Count	Basis of the check
Koha bibliographic records	200	Rows in biblio
Koha item copies	200	Rows in items
Archived DSpace items	20	Items marked in_archive
PDF files linked to archived items	20	Distinct undeleted bitstreams with PDF MIME type
VuFind indexed records	220	Documents in the Solr biblio core
Qdrant evidence points	817	Exact count in kdai_evidence_v1

The case-study evidence consists of the inspected public source revision, author-supplied server outputs, a complete archive of one paired pilot, the two repository PDFs used in that pilot, selected Koha MARC exports, and the completed author review. The public revision used for source inspection is identified in the code reference (Jose, 2026b). The current server service and the separate pilot runner have their own recorded hashes. They are not assumed to be identical to the published revision.

These materials support checks of the implementation and the four pilot sources. They do not constitute a collection-wide extraction audit, a patron study or a production reliability test. The analysis is limited to these recorded observations.

4 Implementation

4.1 Preparing catalogue and repository evidence

The Koha evidence builder extracts a labelled passage from each readable MARC record. The selected fields include title, contributors, subjects, summaries, contents, publication details, ISBNs and some holdings information. This gives the answer service bibliographic context without implying that it has read the book. An answer can use a contents note to identify a relevant chapter or topic, but the note does not substitute for that chapter’s text.

The inspected builder prefers the identifier in MARC 999$c and otherwise falls back to a readable-record counter. Some call-number extraction can also include branch values. These shortcuts require checking before the generated identifiers and holdings labels can be relied on throughout the collection. The two Koha records used in the pilot were checked individually.

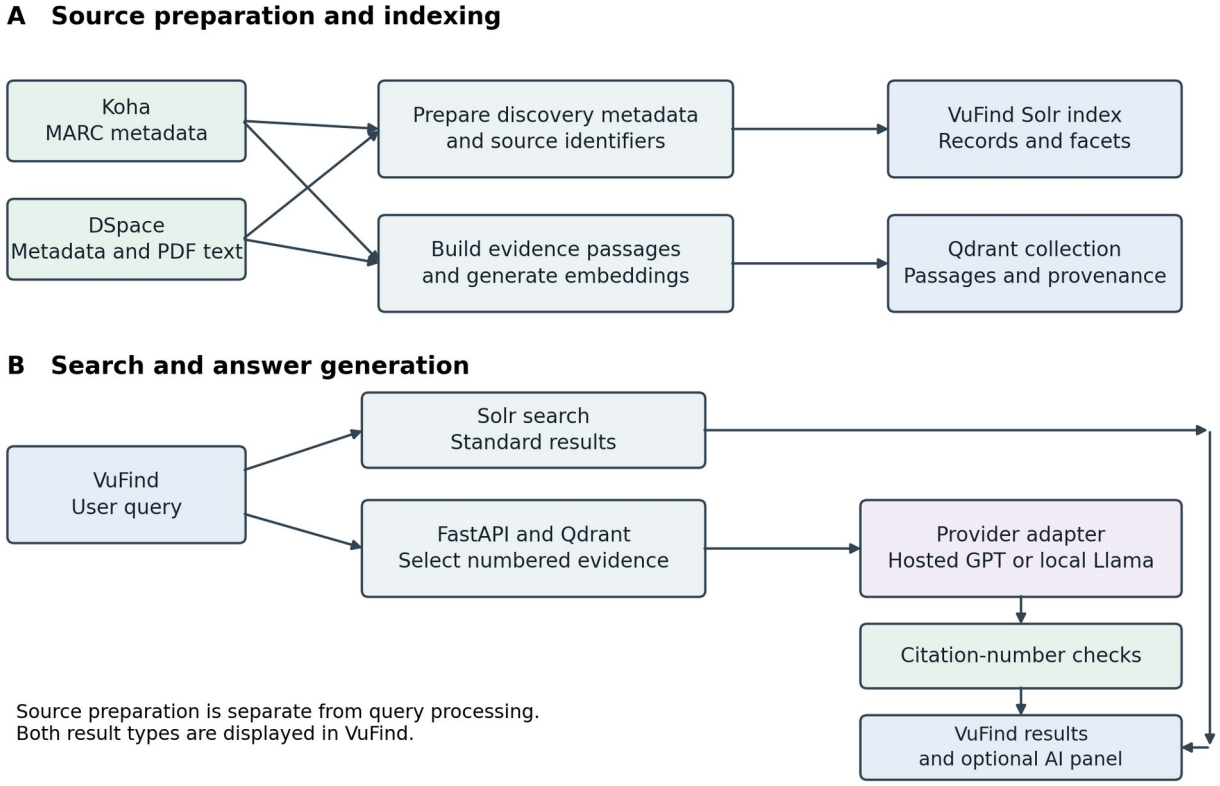
Repository evidence is prepared from metadata and PDF files. The builder uses pypdf to extract pages and divides the text into overlapping passages. Its documented defaults are 220 words per passage with a 40-word overlap. Title, contributor, subject and abstract metadata are prefixed to the extracted material. Because that abstract may describe the whole article, a displayed page label does not by itself prove that every supplied statement occurs on that page.

The builder also derives prototype repository handles from directory order. The pilot’s two handles were reconciled against DSpace bitstream checksums, rather than inferred from staging-folder numbers. That check was necessary because the folders containing the supplied PDFs were numbered item_002 and item_003, while their imported handles ended in /4 and /5.

4.2 Conventional discovery and the optional answer service

VuFind provides the shared search interface. Conventional searches query its Solr index and return records and facets. An optional answer route sends the question to a FastAPI service, which retrieves numbered evidence from Qdrant and passes it to a selected model provider. Figure 1 separates preparation of the indexes from processing a patron’s question.

Figure 1. Source preparation and query processing in the prototype.



The figure represents logical application paths. It does not establish that all PDF text is indexed in Solr or that conventional search has passed fault-isolation testing.

The configured embedding model is BAAI/bge-small-en-v1.5. The service requests 30 vector candidates, removes scores below 0.58, and selects no more than four passages, with at most two per document. Selection first gives up to two qualifying candidates from each source type an opportunity to appear, then fills remaining places. A mixed Koha–DSpace result is therefore influenced by this explicit source-diversity policy.

Context limits also affect what reaches the model. The inspected public code truncates evidence to 500 characters per passage, whereas the earlier preprint records a historical setting of 1,800 characters. The pilot snapshot preserves its actual supplied text. These configurations must be distinguished when interpreting an answer; a model cannot use a relevant passage that has been removed by truncation.

4.3 Model adapters and answer checks

The hosted provider is configured as gpt-6-luna. The local provider is llama3.1:8b through Ollama; its reported configuration has 8.0 billion parameters, Q4_K_M quantisation, a displayed model ID of 46e0c10c039e and 4.9 GB on disk. The model metadata advertises a 131,072-token context capacity, while the pilot requested a local context of 4,096 tokens. The recorded provider aliases do not establish an immutable hosted model revision or a complete local model digest.

The generation instructions require supplied evidence, numerical citations such as [1], and a short answer. They explicitly distinguish catalogue metadata from unread full text and tell the model to treat retrieved instructions as untrusted material. When evidence is insufficient, the requested response is “I could not find sufficient supporting evidence in the connected library collections.” The production prompt contains mixed

guidance about paragraph-end and sentence-end citations; the separate pilot used one explicit rule requiring a citation after each substantive sentence.

The production validator accepts the exact insufficient-evidence response. For other answers it requires at least one bracketed integer and checks that every parsed number falls within the supplied source range. Invalid labels cause an HTTP 502 response before the usual answer is returned. Retrieval with no qualifying passages produces HTTP 404. These conditions are different from a model choosing to abstain. The checks establish citation-number validity; they do not establish that each claim is supported.

The hosted adapter supplies instructions separately from the question and evidence, while Ollama receives a combined prompt. Both providers can therefore use equivalent text without sharing the same internal inference process. The current service defaults allow 600 output tokens for the hosted adapter and 200 for the local adapter; the inspected public revision instead uses 120 locally. The pilot’s separate runner requested 600 for each provider and retained the actual completion metadata.

4.4 Access boundaries

Credentials were read within the server process and excluded from the shared diagnostic outputs. The supplied socket listing shows Qdrant and Ollama listening on loopback addresses. The experiment uses collection evidence and does not require patron identities or borrowing histories. These are specific design and deployment observations, rather than evidence of a complete security assessment. Access controls, rate limits, document rights and failure behaviour still require operational testing before institutional use.

5 Illustrative paired pilot

5.1 Question and saved evidence

The pilot question was “Find catalogue books and repository papers about digital libraries and information access.” It asks for resources from both systems. It does not require a substantive synthesis of arguments across full texts. Retrieval at approximately 21:08 UTC on 29 October selected the four sources shown in Table 2. The saved snapshot retained their order, metadata and complete evidence string.

Table 2. Sources supplied to both providers in pilot P01.

Source	Retrieved resource	Locator	Score
1 DSpace	Digital Library Services: A Comprehensive Review of Usage, Benefits, Challenges, And Future Trends	PDF page 2	0.7658
2 DSpace	Tasks and Functions of Electronic Libraries	PDF page 1	0.7652
3 Koha	Information for a new age: redefining the librarian	Record 38	0.7498
4 Koha	Libraries in the information age: an introduction and career exploration	Record 33	0.7373

A separate runner, kdai_paired_pilot_v1.py, replayed this snapshot without changing the production service. It wrote a pending attempt before each provider request and preserved the original response even when citation validation failed. Ollama ran first, followed by the hosted provider, with no retrieval between them. Both attempts reference the same evidence and instruction checksums. One response per provider was collected.

Each request asked for one paragraph of no more than 70 words and allowed up to 600 output tokens. Local options specified four threads, num_ctx=4096 and keep_alive=5m. No common temperature or seed was

explicitly requested. The procedure holds the supplied evidence constant, but it does not equalise tokenisation, provider infrastructure or all decoding conditions. The pilot snapshot records a separate retrieval interval of 3.354 seconds.

5.2 Checking the original sources

Both saved repository passages were checked against the supplied original PDFs. After Unicode NFKC and whitespace normalisation, each complete passage was an exact substring of its stated PDF page. Source 1 matches PDF page 2, printed page 441; source 2 matches PDF page 1, printed page 185. Both documents identify themselves as journal articles. These checks establish passage presence and page location, rather than the scholarly reliability of every statement in the articles.

A read-only database query associated DSpace handle 123456789/5 with the stored MD5 of the supplied Digital Library Services PDF, and handle 123456789/4 with that of Tasks and Functions of Electronic Libraries. Selected MARC exports for Koha records 38 and 33 agreed with the saved titles, ISBNs and contributors. Record 33 also confirmed the contents note about library careers. Record 38 had a corporate contributor in MARC 710 despite a NULL author value in the simplified biblio table.

The source checks used supplied files and author-supplied terminal outputs, with AI assistance for extraction and comparison. They covered the four pilot sources, not every item in the collection. The DSpace reconciliation compared stored checksum fields; it did not fetch and test every file through the public interface. The original archive hashes and source mappings are retained in Appendix A.

5.3 Observed answers and timings

Table 3. Observations from one saved-evidence request per provider.

Observation	Hosted gpt-6-luna	Local Llama 3.1 8B
Adapter elapsed time	10.413 seconds	238.344 seconds
Whitespace word count	53	81
Maximum 70 words	Passed	Failed
Required numerical labels	Passed; [1][2] and [3][4]	Failed; [SOURCE n]
Provider completion	Completed; no incomplete details	Normal stop
Reported input tokens	1,049	1,054
Reported output tokens	311, including 227 reasoning tokens	101

The hosted answer listed the two repository papers with [1][2], followed by the two catalogue books with [3][4]. It preserved the resource identities and complied with the requested length and numerical-label format. Its usefulness for the topic remains a relevance judgement: the Koha records establish general library-science subjects, not the books’ full-text coverage of digital libraries.

The local answer used [SOURCE 1] through [SOURCE 4], which the integer-only validator cannot parse, and exceeded the word limit. It also called Digital Library Services a book, although the original PDF identifies a journal article. Other parts of the answer were supported: source 2 discusses democratizing knowledge and access to digital resources, and source 4’s contents note mentions library job opportunities. The answer therefore combines supported content with a classification error and failures to follow the output instructions.

Both providers reported normal completion. Ollama stopped after 101 generated tokens, below the allowed ceiling; the hosted response reported no incomplete details. The observed failures cannot therefore be attributed to an output-limit truncation shown in these records. The hosted token total includes 227 reasoning tokens and is not directly comparable with Ollama's visible generation count as a measure of equivalent work.

Ollama reported 33.851 seconds for model loading, 167.052 seconds for prompt evaluation and 37.405 seconds for generation. Loading and input processing account for most of this local request. The hosted adapter interval also included lazy initialisation of local resources before the API call, so it is not a pure remote-inference measurement. Fixed provider order, different infrastructure and a single observation prevent a general speed ranking. Hosted monetary cost, memory peaks and CPU utilisation were not measured.

5.4 Author assessment

I subsequently assessed both saved answers using the review pack and recorded the results on 29 October 2026. The form used 0–4 scales for relevance and completeness, with categorical judgements for source support, source types, citation format and usefulness. I had already seen the model identities and earlier findings. The scoring anchors were also prepared after the pilot outputs were available, making this a retrospective open-label author assessment.

Table 4. Author assessment of the two P01 answers.

Assessment item	Hosted gpt-6-luna	Local Llama 3.1 8B
Relevance 0 to 4	4	3
Completeness 0 to 4	4	3
Claims supported by supplied evidence	Yes	Partly
Source types described accurately	Yes	No
Citation labels follow requested format	Yes	No
Cited sources support associated claims	Yes	Yes
Useful starting point for this patron	Yes	Partly

My notes valued the hosted response's concise separation of repository papers and catalogue books. For the local response, they identified the incorrect label format and the description of source 1 as a book, while crediting its supported summary of electronic libraries and access to knowledge. The form records Yes for source support of cited claims in both answers, alongside Partly for the local answer's overall support and No for its source classification. Those judgements are retained as submitted; no claim-level accuracy percentage was derived from them.

This review adds the developer's assessment of the saved outputs. There was no independent reviewer or agreement measurement. The scores help describe this pilot and its perceived usefulness, but they do not establish comparative performance across questions, collections or institutions.

6 Implementation lessons

Preserving the failed local answer proved useful. The production path can reject an answer with invalid numerical citations before it reaches the normal result or log. The separate runner retained that response, making it possible to distinguish an output-format failure from an empty provider response or an unsupported

answer. For library systems research, failure records are part of the evidence needed to understand what users may encounter.

The source checks also exposed why record identity needs more than a plausible title. A directory number did not equal an imported repository handle, and a NULL value in Koha's simplified author field did not mean that the MARC record lacked a contributor. Checking the underlying fields and file identifiers resolved these two pilot mappings. Similar checks are needed before expanding the collection or relying on holdings and availability labels.

Citation handling needs two separate judgements. A parser can determine whether [1] refers to an existing supplied source. It cannot determine whether the source supports the adjacent sentence. Conversely, a label such as [SOURCE 2] can fail the parser while still pointing to text that supports a particular statement. The service should preserve both the original answer and the validation result so that these outcomes remain distinguishable.

The source-diversity rule and context budget deserve as much attention as the choice of model. Reserving places for both Koha and DSpace may help a mixed-resource question, but it could displace a stronger passage when only one collection is relevant. Truncating a passage after repeated metadata can remove the document text needed to answer the question. These are retrieval and preparation decisions that affect both providers before generation begins.

The selectable adapter design made it possible to send the same saved evidence to hosted and local providers. The pilot also showed the limits of interpreting their operational differences: local loading time, prompt processing and hosted API execution occur under different conditions. A library deciding how to deploy the service would need repeated measurements, expected query volume, measured infrastructure requirements and actual provider charges.

7 Limitations and future work

This case concerns one prototype and a small purpose-built collection. The recorded counts were obtained sequentially and do not form an atomic frozen corpus. Qdrant point totals do not prove coverage or uniqueness, and the DSpace PDF count does not establish public accessibility or successful extraction. The source verification covered two repository passages and selected fields in two catalogue records. Collection-wide mapping, extraction quality and rights remain to be audited.

The illustrative pilot contains one resource-finding question and one answer from each provider. It does not test multi-document synthesis, appropriate abstention across unsupported questions, user satisfaction or long-term reliability. The author assessment is vulnerable to prior knowledge and developer expectations. No general accuracy, cost-effectiveness, privacy-compliance or production-readiness claim follows from these observations.

The next evaluation should preserve a corpus manifest, effective retrieval settings and a held-out set of questions with checked source evidence. Evidence should be retrieved once per question and replayed across providers, with repeated runs and retained failures. Independent librarians could assess claim support, citation correctness and usefulness using a rubric fixed before scoring. Provider order, loading state and response timing should be recorded separately from retrieval. Cost analysis should use actual usage and dated applicable prices.

Further operational work includes testing record links, handling empty retrieval and provider failure, checking access restrictions and documenting how ordinary discovery behaves when the answer service is unavailable.

Any later study with patrons will also need an appropriate ethics and consent assessment. These tasks extend the implementation described here; they are not completed findings of this case study.

8 Conclusion

The prototype connects catalogue metadata and repository evidence within a VuFind discovery interface while providing an optional route to hosted or local generation. The case documents how the evidence was prepared, selected, labelled and passed to the providers, and how a separate runner preserved an answer rejected by the citation-format check.

The paired pilot illustrates a practical distinction between a supported resource recommendation, correct citation labels and accurate source classification. The hosted answer complied with the requested format in this instance; the local answer retained useful content while failing the label and length requirements and misclassifying an article. Preserving the exact evidence and checking the original sources made those differences inspectable. The main contribution is the implementation record and the lessons it provides for further library discovery work, with broader effectiveness still requiring independent evaluation.

Data and code availability

The earlier project preprint is available at https://doi.org/10.5281/zenodo.23532168 (Jose, 2026a). Inspected public code is available at https://github.com/Ajith-jose-lis/koha-dspace-ai-discovery/tree/342b096bde50d317e990a161df05c01ff5025f8e (Jose, 2026b). The pilot archive contains snapshot.json, openai.json, ollama.json and the separate runner. A second archive contains the two source PDFs and their Dublin Core metadata. These archives, the terminal outputs and the completed author form were supplied for private inspection and have not been publicly deposited with this manuscript. Appendix A records the principal checksums and mappings. The original generation logs have not been rewritten to insert the later author assessment.

AI assistance disclosure

I used ChatGPT to assist with prototype software development, manuscript organisation and language editing, examination of supplied evidence and public code, reference formatting and preparation of the architecture figure. Automated extraction and comparison supported the technical source checks. The author assessment in Table 4 records my submitted judgements and is identified separately from those automated checks. I remain responsible for the research decisions, interpretation, references and final manuscript.

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Appendix A Provenance of the illustrative pilot

The pilot evidence was prepared on 29 October 2026. The saved-evidence run called Ollama from 21:10:32.152132 to 21:14:30.496334 UTC, followed by the hosted provider from 21:14:30.509238 to 21:14:40.923487 UTC. Recomputed evidence and instruction checksums agree across both attempt records. Raw answer text, usage, word counts and parsed citations agree with the reported pilot observations. These checks establish internal consistency and transfer integrity, rather than independent provider authentication.

Table 5. Principal pilot and source checksums.

Object	SHA256
Pilot archive	8116365dbac813c0f829bf1c0c7ae96055d7a0cedc9881c56a91ca1a3b489980
Pilot runner	edeb4ea08ec08eb2ea9f78d11ad9cd258e5fd227a2394d0ad719a927aafeb59d
Saved evidence	96f27ca80403ebf8b28ef5168e2a90f9d0b028da94cbbd8f37ce4474ce709ea4
Shared instruction	2ea6d4bfcf88659e12ec7cc163544f0d8340a9c200e1f8315bf09644487269c5
Source PDF archive	19a3fab1399d12ee0e74be36cbcf5d4244bfc0e5fab196abe9f5abef3f4f65f6
Source 1 PDF	a63a1a1c37169523cd19b18e456ba37833ead30317a9b641215e974df59cadd8
Source 2 PDF	6f0bce03472fbfa940cf2bee29ad037081694ff536acc79d62e41340aa6460d8

Source 1 maps to DSpace handle 123456789/5 and bitstream 9f8733cd-88d0-4574-91f4-0b54ebffde7c; its computed MD5 matches the stored value a13c3fb7ead74219b71b053ad2f06212. Source 2 maps to handle 123456789/4 and bitstream 2217e518-759f-4b51-9e58-fefeabf3e529, with matching MD5 a4857451cdce69c8d60dac859400853f. MD5 is retained for reconciliation with existing repository fields; SHA256 identifies the supplied files in the research record.

Sources 3 and 4 map to Koha biblionumbers 38 and 33 respectively. Checked fields were MARC 020, 245 and 710 for record 38, and MARC 020, 100, 245, 505 and 700 for record 33. The snapshot uses VuFind record paths /vufind/Record/dspace-123456789-5, /vufind/Record/dspace-123456789-4, /vufind/Record/38 and /vufind/Record/33. Their browser behaviour was not tested in this source check.

The current server service has recorded SHA256 cce7983c3ba262aea5072ab8fee65c5768348a203709fa8b96a8c3d1e65a8afb. The completed author form is dated 29 October 2026 and records prior exposure to the findings. The original attempt files retain null human_assessment fields because the form was collected afterwards and is held separately.